



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS/
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

TECHNICAL SCIENCES P2/TEGNIESE WETENSKAPPE V2

MAY/JUNE/MEI/JUNIE 2026

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 75

**These marking guidelines consist of 7 pages./
*Hierdie nasienriglyne bestaan uit 7 bladsye.***

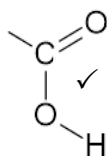
QUESTION/VRAAG 1

- 1.1 C ✓✓ (2)
- 1.2 A ✓✓ (2)
- 1.3 B ✓✓ (2)
- 1.4 D ✓✓ (2)
- 1.5 A ✓✓ (2)

[10]**QUESTION/VRAAG 2**

- 2.1 A molecule containing carbon atoms. ✓✓ /'n Molekuul wat koolstofatome bevat. (2)
- 2.2 Alcohols ✓ /Alkohole (1)
- 2.3 Formyl (group) ✓ /Formiel(groep) (1)

2.4



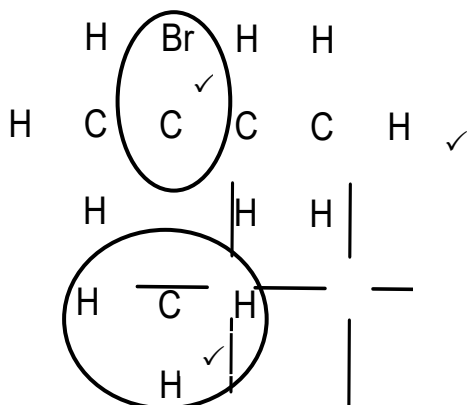
(1)

2.5 $C_nH_{2n}O$ ✓

(1)

2.6.1 2-bromopentane ✓ /2-bromopentaan (2)

2.6.2

**Marking criteria/Nasienkriteria:**

- 1 mark for correct functional group/1 punt vir korrekte funksionele groep
- 1 mark for alkyl group/1 punt vir alkielgroep
- 1 mark for the whole structure correct/1 punt vir volledige struktuur korrek

(3)

2.6.3 Organic molecules that have the same molecular formula, ✓ but different types of chains. ✓ /Organiese molekule met dieselfde molekulêre formule, maar verskillende tipes kettings.

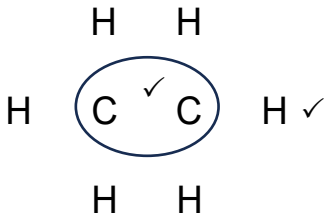
(2)

[13]

QUESTION/VRAAG 3

- 3.1 A series of organic compounds that can be described by the same general formula ✓ and where each member differs from the next by the $-CH_2$ group. ✓ / 'n Reeks organiese verbindings wat deur dieselfde algemene formule beskryf kan word en waarin die een lid van die volgende verskil met die $-CH_2$ -groep. (2)
- 3.2 They contain only single covalent bonds ✓ between carbon atoms. ✓ / Hulle besit slegs enkel kovalente verbindings tussen die koolstofatome. (2)
- 3.3.1 $68,7\text{ }^{\circ}\text{C}$ ✓ (1)
- 3.3.2 $-1\text{ }^{\circ}\text{C}$ ✓ (1)
- 3.4.1
- Pentan-1-ol has both hydrogen bonds and London forces/induced dipole forces/dispersion forces, whereas pentane only has London forces/induced dipole forces/dispersion forces. ✓ / Pentaan-1-ol het beide waterstofbindings en London-kragte/geïnduseerde dipoolkragte/dispersiekragte, terwyl pentaan slegs London-kragte/geïnduseerde dipoolkragte/dispersiekragte het.
 - Pentan-1-ol has stronger intermolecular forces than pentane. ✓ / Pentaan-1-ol het sterker intermolekulêre kragte as pentaan.
- OR/OF**
- Pentane has weaker intermolecular forces than pentan-1-ol. / Pentaan het swakker intermolekulêre kragte as pentaan-1-ol. (2)
- 3.4.2 Higher than ✓ / Hoër as (1)
- [9]**

QUESTION/VRAAG 4

- 4.1.1 Combustion ✓ / Verbranding **OR/OF** oxidation/oksidasie (1)
- 4.1.2 Water ✓ / Water (1)
- 4.2.1 Hydrogenation ✓ / Hidrogenering/Hidrogenasie (1)
- 4.2.2
- 

Marking criteria/Nasienkriteria:

 - 1 mark for correct functional group/1 punt vir korrekte funksionele groep
 - 1 mark for whole structure correct/1 punt vir volledige struktuur korrek
- (2)
- 4.2.3 Nickel ✓ / Platinum/Palladium / Nikkel/Platinum/Palladium (1)
- 4.3.1 $Br_2(g)$ ✓ (1)
- 4.3.2 UV light/UV lig ✓ **OR/OF** Ultraviolet light /Ultraviolet lig (1)

4.4.1 Small organic molecules that can be covalently bonded to each other ✓ in a repeating pattern. ✓ / *Klein organiese molekule wat kovalent aan mekaar gebind kan word in 'n herhalende patroon.* (2)

4.4.2

- (Manufacturing of) squeeze bottles ✓ / *(Vervaardiging van) drukbottels*
- Bullet proof vests ✓ / *Koeëlvaste baadjies*
- (To make) plastic products / *(Om) plastiekprodukte (te maak)*

(Any correct two / *Enige korrekte twee*) (2)

[12]**QUESTION/VRAAG 5**

5.1 Material that has electrical conductivity between that of a conductor and an insulator. ✓✓ / *Materiaal met 'n elektriese geleidingsvermoë tussen dié van 'n geleier en 'n isolator.* (2)

5.2

Feature/ Eienskap	Forward bias/ Meevoorspannend	Reverse bias/ Teenvoorspannend
Connection/ Verbinding	Anode/p-type connected to positive terminal of battery. / <i>Anode/p-tipe verbind aan positiewe terminaal van battery.</i> OR/OF Cathode/n-type connected to negative negative terminal of battery. / <i>Katode/n-tipe verbind aan negatiewe terminaal van battery</i>	Anode/p-type connected to negative terminal of battery. / <i>Anode/p-tipe verbind aan negatiewe terminaal van battery.</i> OR/OF Cathode/n-type connected to positive terminal of battery. / <i>Katode/n-tipe verbind aan positiewe terminaal van battery</i>
Depletion region/ Uitputtingsarea	Narrower / <i>Nouer</i>	Wider / <i>Wyer</i>
Resistance/ Weerstand	Low / <i>Laag</i>	High / <i>Hoog</i>
Conductivity/ Geleiding	Higher (current flow) / <i>Hoër (stroomvloei)</i>	Lower (very small diffusion current) / <i>Laer (baie klein diffusiestroom).</i>

NOTE/LET WEL: One mark for stating each correct difference. / *Een punt vir die stelling van elke korrekte verskil.*

(4)
[6]

QUESTION/VRAAG 6

6.1 Electrical (energy) to chemical (energy). ✓✓/Elektriese (energie) na chemiese (energie). (2)

6.2 Cl^- ✓ **OR/OF** Chloride ion/Chloriedioon (1)

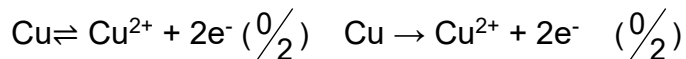
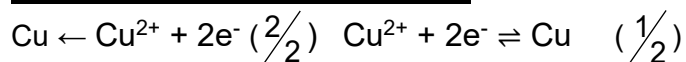
6.3 **NEGATIVE MARKING FROM QUESTION 6.2/NEGATIEWE NASIEN VANAF VRAAG 6.2**

$\text{Cl}^-(\text{aq})$ loses electrons/is oxidised ✓✓ (to form Cl_2). / $\text{Cl}^-(\text{aq})$ verloor elektrone/ word geoksideer om (Cl_2 te vorm). (2)

6.4 Intensity of (blue) colour decreases ✓ (as copper forms at the cathode). / Intensiteit van (blou) kleur neem af (soos koper vorm by die katode). (1)

6.5 $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ ✓✓

Marking criteria/Nasienkriteria:



NOTE/LET WEL: Do not penalise if the phases are omitted. / Moenie penaliseer indien fases weggelaat is nie.

(2)

6.6 $\text{Cu}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \checkmark \rightarrow \text{Cl}_2(\text{g}) + \text{Cu}(\text{s}) \checkmark$ ✓ bal

Marking criteria/Nasienkriteria:

- 1 mark for correct reactants/1 punt vir korrekte reaktante
- 1 mark for correct products/1 punt vir korrekte produkte
- 1 mark for balancing/1 punt vir balansering

NOTE/LET WEL: Do not penalise if the phases are omitted. / Moenie penaliseer indien fases weggelaat is nie.

(3)

6.7

- Noisy while in operation ✓/Lawaaierig tydens werking
- Landscapes become unsightly/Landskappe word onooglik
- Wind farms disrupt habitats/Windplase ontwrig habitate
- Harmful to wildlife (such as turbines killing birds)/Skadelik vir wildlewe (soos turbines wat voëls doodmaak)
- Inconsistent and unpredictable because of fluctuating wind speed/Inkonsekwent en onvoorspelbaar as gevolg van wisselende windspoed

(Any correct one/Enige korrekte een) (1)

6.8

- Nuclear (energy) ✓/Kern(energie)
- Biodiesel/Biodiesel
- Fuel cell/Brandstofsel
- Hydroelectric (energy)/Hidroelektriese (energie)
- Solar (energy)/Stralings(energie)

(Any correct one/Enige korrekte een) (1)

[13]

QUESTION/VRAAG 7

- 7.1
- Concentration/*Konsentrasie*: $1 \text{ mol} \cdot \text{dm}^{-3}$ ✓
 - Temperature/*Temperatuur*: 25°C or 298 K

(Any one/*Enige een*) (1)

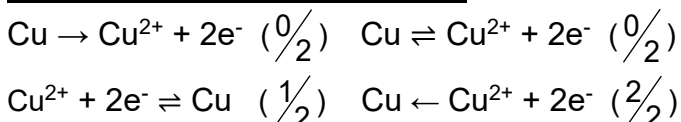
- 7.2
- $$E^\theta_{\text{cell/sep}} = E^\theta_{\text{cathode/katode}} - E^\theta_{\text{anode/anode}}$$
- $$E^\theta_{\text{cell/sep}} = E^\theta_{\text{reduction/reduksie}} - E^\theta_{\text{oxidation/oksidasie}}$$
- $$E^\theta_{\text{cell/sep}} = E^\theta_{\text{oxidising agent/oksideermiddel}} - E^\theta_{\text{reducing agent/reduseermiddel}}$$
- $$= +0,34 \checkmark - (-0,76) \checkmark$$
- $$= 1,10 \text{ V} \checkmark$$

✓ Any one/
Enige een**Marking criteria/Nasienkriteria:**

- 1 mark for formula/1 punt vir formule
- 1 mark for EACH substitution/1 punt vir ELKE substitusie
- 1 mark for answer with unit/1 punt vir antwoord met eenheid

(4)

- 7.3 $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ ✓✓

Marking criteria/Nasienkriteria:

NOTE/LET WEL: Do not penalise if the phases are omitted./Moenie penaliseer indien fases weggelaat word nie.

(2)

- 7.4.1 **Y** ✓

(1)

7.4.2 **NEGATIVE MARKING FROM QUESTION 7.4.1/NEGATIEWE NASIEN VANAF VRAAG 7.4.1**

- Electrode **Y** is losing electrons.✓✓ /Electrode **Y** is the negative electrode./Electrode **Y** is oxidised. /*Elektrode Y verloor elektrone./Elektrode Y is die negatiewe elektrode./Elektrode Y word geoksideer.*
- Electrons flow from the anode/**Y**/Zink to the cathode/**X**/copper./*Elektrone vloei vanaf die anode/Y/Sink na die katode/X/koper.*

(Any correct one/*Enige korrekte een*) (2)

- 7.5 • There will be no movement of ions through the salt bridge. ✓✓ /Daar sal geen beweging van die ione deur die soutbrug wees nie.

OR/OF

- The neutrality of the electrolytes in the half-cells will not be maintained./Die neutraliteit van die elektroliete in die halfselle sal nie behou word nie.

OR/OF

- The circuit will be incomplete./Die stroombaan sal onvolledig wees.

OR/OF

- The net charge of the electrolyte in the half-cells will increase./Die netto lading van die elektroliet in die halfselle sal toeneem.

OR/OF

- The half-cell reactions will slow down and stop./Die halfselreaksies sal stadiger word en stop.

(2)
[12]

TOTAL/TOTAAL: 75